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Probability and Information Theory

A quick and dirty introduction

Reminder: What is probability?

1. **Incomplete modeling** (JO ABHI JAAN-NA NAHI CHAHTE) : In principle, you can use aerodynamics to calculate which way a coin will land by accounting for wind etc but in practice we take the outcome to be random.
2. **Incomplete observability** (JO ABHI NAHI JAANTE) : Buying a second hand car / bike / phone.
3. **Inherent stochasticity in the system being modeled** (JO JAANA NAHI JAA SAKTA) : Dynamics of subatomic particles seen as probabilistic in most quantum mechanics interpretations. Has similarities with incomplete modeling.

Reminder: Probability terms you should remember

- Random experiment
- Outcomes
- Sample Space
- Events
- Probability of an event

Reminder: Probability terms you should remember

- Random experiment: Rolling a fair 6-sided dice.
- Outcomes: 1, 2, 3, 4, 5, 6.
- Sample Space: Collection of all outcomes $\{1, 2, 3, 4, 5, 6\}$.
- Events: Getting 1 ($\{1\}$), Getting an even number ($\{2, 4, 6\}$) etc.
- Probability of an event: A measure of the likelihood of an event occurring. Takes values from 0 to 1. For example, $P(\{3, 6\}) = \frac{1}{2}$.

Example: Ek Ajnabee

- You're at a friend's party and start a random conversation with a stranger. You find them really interesting to talk to, and wonder when their birthday could be.
- Is this partial observability or incomplete modeling?
- What is the sample space? How many possible outcomes?
- Ignoring 29 Feb and assuming uniform birth rates, what probability distribution would you assign to their birthday?

Reminder: Probability terms you should remember

- Random Variable
- Joint Probability
- Conditional Probability
- Bayes Theorem

Reminder: Probability terms you should remember

- Random Variable: A number associated with each outcome of a random experiment. Example: Number of dots on a fair dice.
- Joint Probability: The probability of events A and B both occurring / probability of “A and B”. They could originally come from different sample spaces.
- Conditional Probability: The probability of A given B is the probability of A taking into account the information that B has already happened.

Joint and Conditional Probability

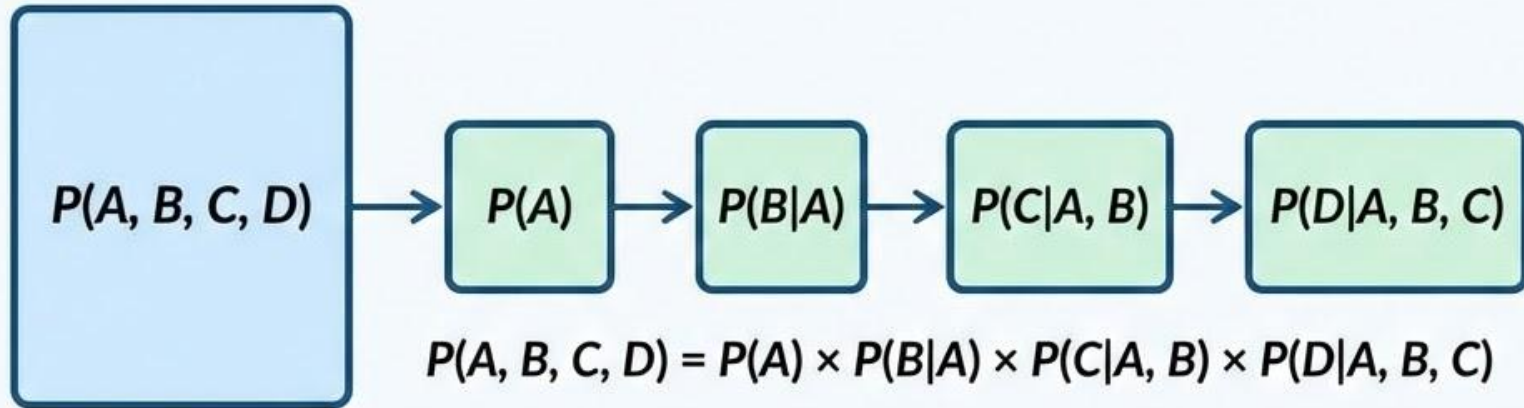
$$P(X, Y) = P(X \text{ and } Y)$$

$$\mathbf{P}_{XY} = \begin{bmatrix} P(x_1, y_1) & P(x_1, y_2) \\ P(x_2, y_1) & P(x_2, y_2) \end{bmatrix}$$

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}$$

Chain rule of conditional probability

Joint probabilities can be decomposed into a product of several conditional probabilities.



Learning from Evidence: Bayes' Theorem

Core Idea: Bayes' Theorem provides a way to update the probability of a hypothesis (A) based on new evidence (B). It allows us to calculate $P(A|B)$ if we know $P(B|A)$.

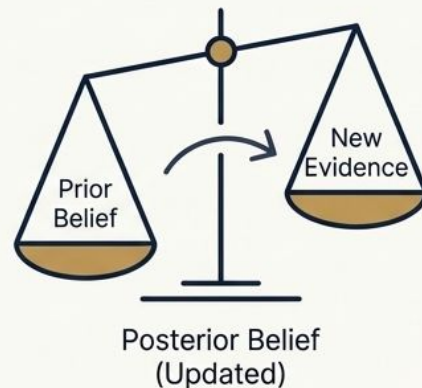
$$\begin{array}{c} \text{Posterior} \leftarrow P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \longrightarrow \text{Prior} \\ \text{(updated probability)} \qquad \qquad \qquad \text{Likelihood (probability of evidence given hypothesis)} \\ \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \text{Evidence (total probability of evidence)} \end{array}$$

In Machine Learning Classification

We want to find $P(\text{class} | \text{features})$. Bayes' Theorem tells us this is proportional to $P(\text{features} | \text{class}) * P(\text{class})$.

$P(\text{class} | \text{features})$ is proportional to $P(\text{features} | \text{class}) * P(\text{class})$

- $P(\text{class} | \text{features})$: Posterior (what we want to find)
- $P(\text{features} | \text{class})$: Likelihood (what the model learns)
- $P(\text{class})$: Prior (our initial belief about class frequencies)



Example: Medical diagnosis with Bayes Theorem

- A person has 5% chance of having a fever, while someone with Typhoid has a 50% chance of having a fever. The natural rate of Typhoid in the population is 2%, what is the probability that someone with a fever is suffering from Typhoid?

Example: Medical diagnosis with Bayes Theorem

- A (Prior of what we're interested in measuring).
 - B (Evidence potentially telling us something about A).
 - $A|B$ (Posterior of what we're interested in measuring).
 - $B|A$ (Likelihood of seeing the evidence if the hypothesis is true).
-
- $P(A) = 2\%$, $P(B) = 5\%$, $P(B|A) = 50\%$
 - $P(A|B) = 0.50 * 0.02 / 0.05 = 20\%$

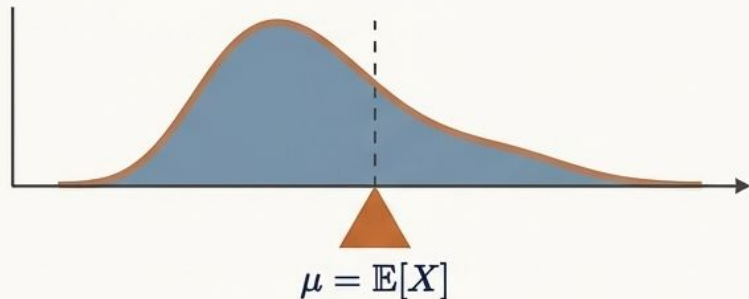
Reminder: Probability terms you should remember

- Expected Value
- Variance
- Covariance
- Correlation

The Tool: Summarizing a Distribution's Shape

$$\mathbb{E}[X]$$

The long-run average; the “center of mass” of the probability distribution.

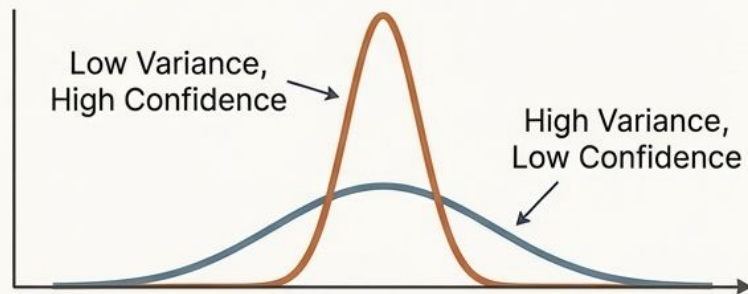


$$\mathbb{E}[X] = \sum_x x \cdot P(X = x)$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$\text{Var}(X)$$

A measure of spread, uncertainty, or “surprise”.



$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

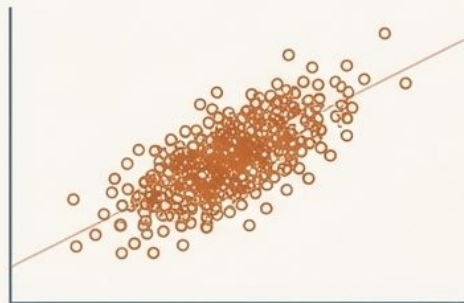
The standard deviation is $\sigma = \sqrt{\text{Var}(X)}$.



ML Connection: We carefully control the variance of initial network weights to ensure stable training. The variance in a Bayesian model's predictions tells us about its confidence.

The Matrix Perspective: How Variables Move Together

Positive Covariance



e.g., Height vs. Weight

Negative Covariance



e.g., Hours Gaming vs. GPA

Zero Covariance



Covariance $\text{Cov}(X, Y)$

A measure of the direction of the linear relationship between two random variables.

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

The Covariance Matrix Σ

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) \end{bmatrix}$$



ML Connection: The covariance matrix is fundamental to understanding the shape of multivariate data. It is the key parameter of a Multivariate Gaussian distribution, used in Principal Component Analysis (PCA) and Gaussian Mixture Models (GMMs).

Two Machine Learning Paradigms for Classification

- *J'adore venir en cours tôt le matin.*

Can you guess what language this is? Can you write “August is truly the best month” in this language?

- मुझे सुबह-सुबह क्लास में आना बहुत पसंद है।

Can you guess what language this is? Can you write “August is truly the best month” in this language?

Two Machine Learning Paradigms for Classification

- Discriminative Models: Directly estimating $P(\text{Class}|\text{Evidence})$ using training data, and then predict the class of new data points based on the estimates.
- Examples: Logistic Regression, SVMs, MLPs
- Discriminative models allow you to make predictions at inference time using patterns without directly understanding the nature of data you are working with.

Two Machine Learning Paradigms for Classification

- Generative Models: Estimating the full joint probability distribution $P(\text{Class}, \text{Evidence})$ of how training data and the associated class labels are distributed.
- Examples: Naive Bayes, HMMs, GANs, VAEs
- In addition to running inference, you can also use generative models to sample data distributed identically to the training data.

LLMs as Generative Models

- Consider the random experiment of seeing the next word in a sentence.
- The sample space is all the words in our vocabulary.
- The probability of each outcome is affected by the previous words in the sentence, we can use the chain rule of conditional probability to calculate the overall probability of the sentence.
- I loaded up the text of “Emma” by Jane Austen as example, let’s see if you can predict the most common words in it!

LLMs as Generative Models

- What do you estimate was the most common word in the book?
- $P(\text{word}) = ?$

to		5239		3.24%
the		5201		3.22%
and		4896		3.03%
of		4291		2.66%
i		3178		1.97%

LLMs as Generative Models

- We're now interested in words that fill the slot "I _____"
- $P(\text{word} \mid \text{"I ____"}) = ?$

I am		395		12.43%
I have		281		8.84%
I do		181		5.70%
I shall		120		3.78%
I was		111		3.49%

LLMs as Generative Models

- We're now interested in words that fill the slot “I have _____”
- $P(\text{word} \mid \text{“I have ____”}) = ?$

I have been	38	13.52%
I have not	37	13.17%
I have no	35	12.46%
I have a	19	6.76%
I have heard	16	5.69%

Example: 20 Questions with Ek Ajnabee

- You're at a friend's party and start a random conversation with a stranger. You find them really interesting to talk to, and wonder when their birthday could be.
- They ask you if you can guess it by asking them yes or no questions. Which of these questions should you start with? Why?
- Is it in January?
- Is it on or before 30 June?
- Is it on or after 15 December?

“Surprise” and “Information”

What is the probability of them saying yes to “Is it in January?”?

It would be slightly surprising if they say yes instead of no to this questions, but if they it's great! Seeing a low probability outcome will give us a lot of information.

But most likely, they will say no, and we will get to know very little new information about their birthday.

“Surprise” and “Information”

Let's take this logic to the extreme. If we ask them “Is it on or before 31 December?”, they will say yes with probability 1 but this answer will tell us nothing.

We're slowly seeing that the rarer the outcome of a random experiment, the higher the information we get on seeing it.

Information in an outcome

We formalize this notion of surprise by defining the entropy of an outcome with probability p_i as $-\log_2(p_i)$.

Consider getting heads on a fair coin, the information of this outcome is $I(H) = -\log_2(1/2) = \log_2(2) = 1$.

We call this amount 1 bit. It represents information that can be confirmed with certainty by a single yes or no question.

Entropy of an experiment

Consider an biased coin with the probability of getting heads as 90%.
Let us calculate the information of getting heads and tails on this coin.

Information in getting Heads $I(H) : -\log_2(0.9) = 0.15$

Information in getting Tails $I(T) : -\log_2(0.1) = 3.32$

But before event flipping this coin, can we calculate the expected value of this information? We call this quantity the entropy of the experiment of flipping this coin.

$$H(X) = 0.9 * 0.15 + 0.1 * 3.32 = 0.47$$

Quantifying Uncertainty: Self-Information and Entropy

Self-Information, $I(x)$

Measures the surprise of a single, specific outcome x .

$$I(x) = -\log P(x)$$

Low probability $P(x) \rightarrow$ High information $I(x)$

Entropy, $H(X)$

Measures the average uncertainty of the entire random variable X . It's the expected value of the self-information over all possible outcomes.

$$H(X) = -\sum P(x) \log P(x)$$

Example: Coin Flips (base-2 log)

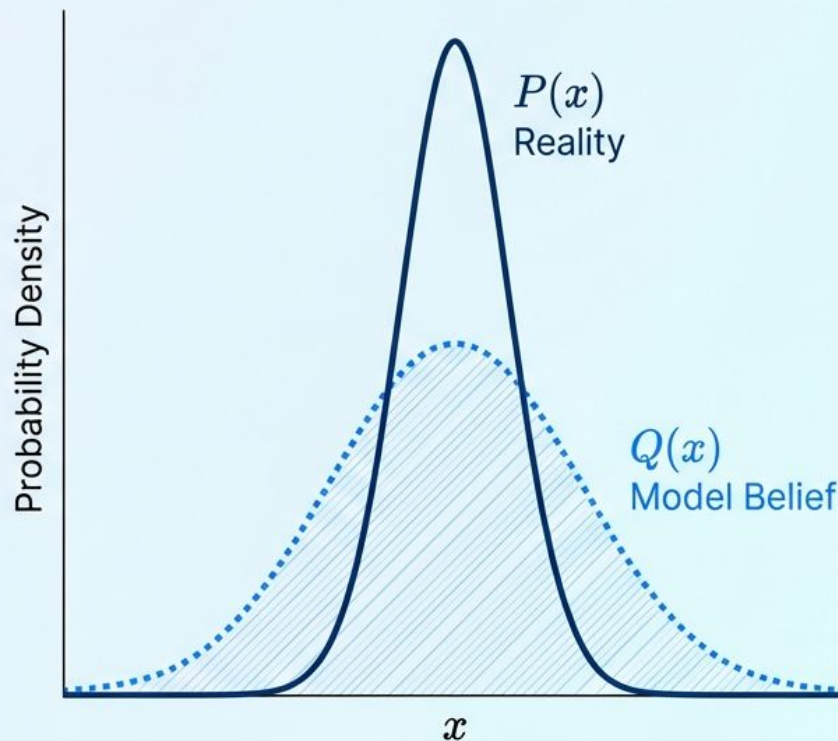
Case	Distribution	Uncertainty Level	Entropy
Fair coin	[0.5, 0.5]	Max uncertainty	1 bit
Biased coin	[0.9, 0.1]	Some certainty	0.47 bits
Certain outcome	[1.0, 0.0]	No uncertainty	0 bits

The Modeling Problem: Perception vs. Reality

In the real world, we rarely know the true distribution P . We estimate it using a model Q .

- $P(x)$ **(Reality)**: The true probability distribution (e.g., a biased coin is actually 90% Heads).
- $Q(x)$ **(The Model)**: Our predicted probability distribution (e.g., we *think* the coin is Fair, 50% Heads).

The Core Question: Events occur according to Reality (P), but we experience surprise according to our Model (Q). How much surprise do we experience on average?



Example: Cross-Entropy

Consider an biased coin with the probability of getting heads as 90%. I give you this coin but I deceive you and say that it a fair coin. This is how you will calculate the surprise:

Information in getting Heads $I(H) : -\log_2(0.5) = 1.0$

Information in getting Tails $I(T) : -\log_2(0.5) = 1.0$

What is the expected value of this information based on the true probabilities of the coin?

$$0.9 * 1.0 + 0.1 * 1.0 = 1.0$$

Example: Cross-Entropy

Consider an biased coin with the probability of getting heads as 90%. I give you this coin but I deceive you and say that it a fair coin.

We know that the true entropy for this coin toss is $H(X) = 0.47$

Since you had the wrong estimate of the coin probability, you were more surprised than required (1.0 bit instead of 0.47 bit).

In general, since our models don't learn the true probability distributions of data, this additional surprise can be used as an error/loss in training the model.

Example: KL Divergence

Consider an biased coin with the probability of getting heads as 90%. I give you this coin but I deceive you and say that it a fair coin.

We know that the true entropy for this coin toss is $H(X) = 0.47$

Since you had the wrong estimate of the coin probability, you were more surprised than required (1.0 bit instead of 0.47 bit).

This additional surprise $1.00 - 0.47 = 0.53$ bit is called the KL divergence.

Defining Cross Entropy

$$H(P, Q) = - \sum_x P(x) \log Q(x)$$

We average over what *actually* happens (Reality)

We calculate surprise based on what we *thought* would happen (Model)

Key Insight: Because our model Q is imperfect, we will always be *more* surprised than necessary.

Inequality: $H(P, Q) \geq H(P)$ (Cross Entropy creates a 'floor' for our error)

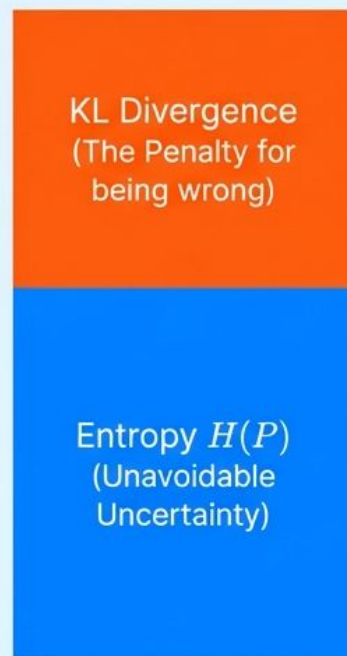
Kullback-Leibler (KL) Divergence

KL Divergence measures the “distance” between two probability distributions. It quantifies the *inefficiency* or “waste” of assuming Q when the truth is P .

$$D_{KL}(P \parallel Q) = H(P, Q) - H(P)$$

KL Divergence = [Total Cross Entropy] - [Inherent Entropy]

Total Surprise ($H(P, Q)$)



We want to
minimize this.

Asymmetry of KL Divergence

$$H(P, Q) = - \sum_{x \in \mathcal{X}} P(x) \log Q(x)$$

$$H(Q, P) = - \sum_{x \in \mathcal{X}} Q(x) \log P(x)$$

$$D_{\text{KL}}(P \parallel Q) = H(P, Q) - H(P)$$

$$D_{\text{KL}}(Q \parallel P) = H(Q, P) - H(Q)$$

Please note that $H(P, Q) \neq H(Q, P)$ and $D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$

Deriving the Loss Function



1. We want to minimize KL Divergence:

$$H(P, Q) - H(P)$$



2. Crucial Observation: $H(P)$ is the Entropy of the training data (the labels).
The training data is fixed. We cannot change reality.



$$\text{Loss} = H(P, Q) - \cancel{H(P)}$$

Constant



4. Conclusion: Minimizing KL Divergence is mathematically identical to minimizing Cross Entropy.

This is why we use Cross Entropy Loss to train Neural Networks.
We simply minimize the surprise of our model against the data.

Cross Entropy in LLMs

Recall the text prediction context: Use the image of the “I have no...” word distribution table.

I have been	38	13.52%
I have not	37	13.17%
I have no	35	12.46%
I have a	19	6.76%
I have heard	16	5.69%

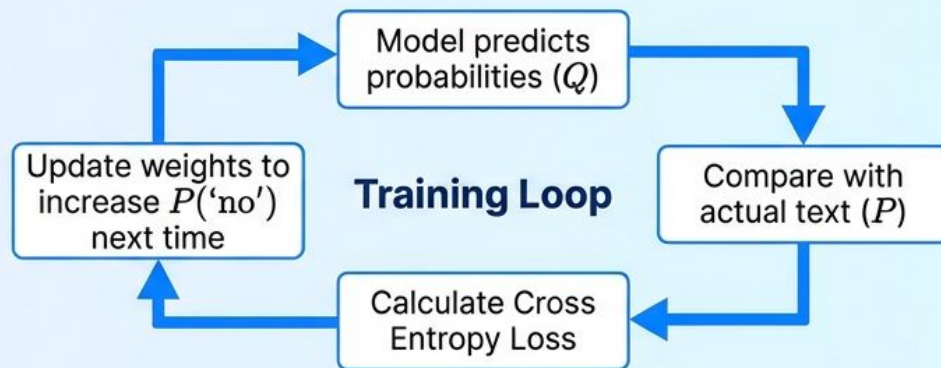
Text explanation:

The ‘True’ Distribution (P):

The actual next word Jane Austen wrote (e.g., ‘I have **no**...’).
 $P(\text{‘no’}) = 1$.

The Model (Q):

The LLM predicts a list of likely words:
{been: 13%, not: 13%, no: 12%...}.



“Training an LLM is literally the process of minimizing the Cross Entropy between the model’s guesses and human text.”

Summary: The Metrics of Uncertainty

Entropy $H(P)$

Nature's Uncertainty

The lower bound of surprise. Defined by the data itself.

Cross Entropy $H(P, Q)$

The Model's Surprise

How surprised we are when observing Nature (P) while using our Model's map (Q).

KL Divergence D_{KL}

The Penalty

The difference between the two ($H(P, Q) - H(P)$).
The information lost due to model error.

In Deep Learning, we minimize Cross Entropy to drive the Model (Q) toward Reality (P).

**Thank you all for a great session please fill the
feedback form.**